4 Non-SI units

It is recognized that some non-SI units are widely used and that this is expected to continue for many years. It is therefore important to recall the values of these non-SI units in terms of SI units, because the SI is the internationally agreed reference with respect to which all other units are defined. A non-exhaustive list of non-SI units is given in Table 8, grouped into indicative unit categories to aid explanation.

Table 8. Non-SI units

Unit category	Quantity	Name of unit	Symbol for unit	Value in SI units
Long-standing units of time and angle	time	minute	min	1 min = 60 s
		hour	h	1 h = 60 min = 3600 s
		day	d	1 d = 24 h = 86 400 s
	plane and phase angle	degree	°	1° = (π/180) rad
		minute	′	1′ = (1/60)° = (π/10 800) rad
		second [a]	″	1″ = (1/60)′ = (π/648 000) rad
Historical names for decimal multiples and submultiples of SI units	area	are [b]	a	1 a = 1 dam^2 = 10^2 m^2
		hectare [b]	ha	1 ha = 1 hm^2 = 10^4 m^2
		barn [c]	b	1 b = 100 fm^2 = 10^{-28} m^2
	volume	litre [d]	l, L	1 l = 1 L = 1 dm^3 = 10^{-3} m^3
	mass	tonne [e]	t	1 t = 1 Mg = 10^3 kg
	length	angstrom [f]	Å	1 Å = 0.1 nm = 10^{-10} m
	acceleration	gal [g]	Gal	1 Gal = 1 cm/s^2 = 10^{-2} m/s^2
	pressure	bar [h]	bar	1 bar = 0.1 MPa = 10^5 Pa
Internationally recognised units that are not decimal multiples or submultiples of SI units	mass	dalton [i]	Da	1 Da = 1.660 539 068 92(52) × 10^{-27} kg
	length	astronomical unit [j]	au	1 au = 149 597 870 700 m
		nautical mile [k]		1 nautical mile = 1852 m
	speed	knot [k]		1 nautical mile per hour = (1852/3600) m/s
	energy	electronvolt [l]	eV	1 eV = 1.602 176 634 × 10^{-19} J
Units used in specialized technical disciplines	logarithmic ratio quantities	neper [m]	Np	[m]
		bel [m]	B	[m]
		decibel [m]	dB	[m]
	reactive power	var [n]	var	1 var = 1 V A = 1 W

(*a*) For some applications such as in astronomy, small angles are measured in arcseconds (with unit symbols: as or ″) or milliarcseconds, microarcseconds and picoarcseconds (with unit symbols: mas, μas and pas, respectively) where arcsecond is an alternative name for second of plane angle.

(*b*) The CIPM adopted the units are and hectare, and their symbols, in 1879. They are used to express land area.

(*c*) The barn is used in nuclear physics to express effective cross sections.

(*d*) The litre and the symbol, lower-case l, were adopted by the CIPM in 1879. The alternative symbol, capital L, was adopted by Resolution 6 of the 16th CGPM in 1979 in order to avoid the risk of confusion between the letter l (el) and the numeral 1 (one).

(*e*) The tonne and its symbol, t, were adopted by the CIPM in 1879. This unit is sometimes referred to as "metric ton" in some English-speaking countries.

(*f*) The angstrom is used to express lengths of chemical bonds, sizes of atoms and molecules, and wavelengths.

(*g*) The gal is used in geodesy and geophysics to express acceleration due to gravity and originates from the CGS system.

(*h*) The bar and its symbol were included in Resolution 7 of the 9th CGPM in 1948.

(*i*) The dalton (symbol Da) and the unified atomic mass unit (symbol u) are alternative names (and symbols) for the same unit, equal to 1/12 of the mass of a free carbon 12 atom, at rest and in its ground state. This value of the dalton is the value recommended in the 2022 CODATA Recommended Values of the Fundamental Physical Constants.

(*j*) Defined by Resolution B2 of the XXVIII General Assembly of the International Astronomical Union in 2012.

(*k*) The nautical mile and the knot are used in marine and aerial navigation to express distance and speed, respectively. There is no internationally agreed unit symbol for either unit. The size of the nautical mile was defined by the 1st Extraordinary International Hydrographic Conference in 1929.

(*l*) The electronvolt is the kinetic energy acquired by an electron in passing through a potential difference of one volt in vacuum. Following adoption of Resolution 1 of the 26th CGPM in 2018, which fixed the numerical value of the elementary charge, this conversion factor has been exact.

(*m*) The neper, bel and decibel are used to express values of specified logarithmic ratio quantities. When using these units, it is important that the quantity be specified, and that any reference value used be specified. The neper, Np, is used to express the values of quantities whose numerical values are based on the use of the neperian (or natural) logarithm, ln = $\log_e$, usually applied to logarithmic amplitude ratios. The statement $L_A = n$ Np (where n is a number) is interpreted to mean that $n = \ln(A_2/A_1)$. The symbol A typically denotes the amplitude of a sinusoidal signal, and L_A is then called the neperian logarithmic amplitude ratio. The bel and the decibel, B and dB, where 1 dB = (1/10) B, are used to express the values of logarithmic ratio quantities whose numerical values are based on the decadic logarithm, lg = $\log_{10}$, usually applied to logarithmic power ratios. The statement $L_X = m$ dB = (m/10) B (where m is a number) is interpreted to mean that $m = 10 \lg(X/X_0)$. If X denotes a mean square signal or power-like quantity, L_X is called a power level referred to X_0.

(*n*) The var is a special name for the unit of reactive power. It is identical to the coherent SI unit volt ampere.